Outcome School

AI and ML Curriculum

AI and Machine Learning Program

Project Based Learning

OPPORTUNITY

This program makes you **job-ready** for roles such as:

- **AI Engineer**
- Agentic AI Engineer
- AI Architect
- LLM Engineer
- Gen AI Engineer
- MLOps Engineer
- Machine Learning Engineer
- Deep Learning Engineer

ELIGIBILITY

Working tech professionals who want to learn AI from scratch and **transition** into high-paying **AI roles** through a strong understanding of AI, Machine Learning, and System Design.

PREREQUISITE

Any programming language. **Even without Python experience**, if you know any **other language**, you can learn Python while building the projects.

DURATION

6 months: The overall duration of the program will be six months, ensuring enough time to cover every topic in depth.

LIVE CLASSES

All live classes will be conducted exclusively on **weekends**, making it easier for working professionals and students to attend without affecting their weekday schedule.

CURRICULUM OUTLINE

- **AI Engineering**
- Tools and Libraries
- Data Analysis and Visualisation
- Machine Learning Fundamentals
- Deep Learning
- Gen AI
- MLOps, LLMOps
- AI and ML System Design
- Reinforcement Learning

CURRICULUM

AI Engineering

- AI Engineering Overview
- LLM Fundamentals
- LLM Internals
- Tokenization
- Positional Encodings
- Q, K, V Matrices
- Attention Mechanism
- Self-Attention and Multi-Head Attention
- Causal Masked Attention
- Transformer Architecture
- KV Cache
- Paged Attention
- Mixture of Experts (MoE)

CURRICULUM

AI Engineering

- Cross Attention
- vLLM
- Prompt Engineering
- Prompt Chaining
- Chain of Thought (CoT) Prompting
- Context Engineering
- RAG
- Vector Databases
- Fine-tuning
- Parameter-Efficient Fine-Tuning (PEFT)
- Low-Rank Adaptation (LoRA), QLoRA
- Quantization and Optimizations
- Model Compression

CURRICULUM

AI Engineering

- Knowledge Distillation
- SLMs and Model Distillation
- ReAct Pattern
- AI Agent, Agentic AI
- Tool use in Agents
- Memory in Agents
- Model Context Protocol (MCP)
- Agent Architecture
- Subagent, Multi-Agent Systems
- Orchestration and Routing
- Evaluation of LLMs and Agents
- LLM as a Judge
- Multimodal AI

BUILD YOUR OWN X

We believe in learning the internals, and the best way to do that is by building things from scratch.

Build your own LLM from scratch

Build your own AI Agent from scratch

Build your own AI Tutor from scratch

Build your own CNN from scratch

Build your own Neural Network from scratch

Gen AI PROJECTS

01

Text Generation - ChatGPT

02

Image Generation - Text to Image Generation App

03

Audio Generation - Music Generation App

Tools and Libraries

- PyTorch
- TensorFlow
- Keras
- Ollama
- Hugging Face
- LangChain
- LangGraph

Data Analysis and Visualisation

- Numpy
- Pandas
- Matplotlib
- Seaborn

Machine Learning Fundamentals

- Machine Learning Overview
- Supervised and Unsupervised Learning
- Self-supervised Learning, Contrastive Learning
- Linear Regression, Logistic Regression
- Gradient Descent
- Maths behind Gradient Descent
- Loss Functions
- Hyperparameter Tuning
- Epoch, Batch, Batch Size, and Iteration
- Embeddings
- Logits
- Cross-Entropy

Deep Learning

- Neural Networks
- Feed Forward Neural Networks (FFNNs)
- Activation Functions
- Backpropagation
- Convolutional Neural Networks (CNN)
- Dropout, Batch Normalization
- Optimizers

Gen AI

- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs)
- Autoregressive Models
- Diffusion Models
- Video Models
- Transformers
- Attention Mechanism
- Large Language Models (LLMs)
- Reasoning Models
- Encoder–Decoder Architecture
- Multimodal Generative Models
- Optimization and Scaling Techniques

MLOps, LLMOps

- Data Management & Versioning
- Model Development & Training
- Evaluation & Testing
- Model Deployment & Serving
- Cloud vs On-device Model Deployment
- Monitoring & Logging
- Infrastructure & Platform

AI and ML System Design

- AI Tutor
- AI Coding Agent
- Multimodal Search System
  - Text
  - Image
  - Video

Reinforcement Learning

- Markov Decision Processes (MDPs)
- Q-Learning
- Policy Gradients
- Exploration vs Exploitation
- Deep Reinforcement Learning
- RLHF (Reinforcement Learning from Human Feedback)

LIVE PROJECT

We have a range of AI product ideas that we can brainstorm together and select one to work on.

Gain hands-on experience by working on a real AI project from ideation to execution.

Showcase a real-world AI project in your resume and portfolio.

Consider this as an AI Internship at a company, making it extremely valuable for your job prospects.

RESEARCH PAPERS

Reading AI research papers helps you understand the latest AI techniques and ideas, making it easier to use or create advanced AI solutions.

Attention Is All You Need

Generative Adversarial Nets

Distilling the Knowledge in a Neural Network

YOUR TEACHER AND MENTOR

Amit Shekhar

Founder @ Outcome School • IIT 2010-14 • I have taught and mentored many developers, and their efforts landed them high-paying tech jobs, helped many tech companies in solving their unique problems, and created many open-source libraries being used by top companies. I am passionate about sharing knowledge through open-source, blogs, and videos.

THANK YOU

Thank you for exploring this curriculum.

At Outcome School, our vision is to make tech education outcome-focused.

Our mission is to provide tech education to the students through project-based learning to achieve the outcome they aspire to.

We're grateful to be part of your journey and look forward to the real-world outcomes you'll create.

Outcome School